

SAKHUNYA, Russian Federation

WMO: 273730

Lat: 57.667N

Lon: 46.633E

Elev: 575

StdP: 14.39

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -19.0	(c) -12.9	(d) -24.1	(e) 1.5	(f) -18.5	(g) -17.7	(h) 2.1	(i) -12.6	(j) 22.4	(k) 20.3	(l) 20.8	(m) 18.7	(n) 5.5	(o) 320	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.1	(c) 83.9	(d) 68.4	(e) 80.6	(f) 66.7	(g) 77.4	(h) 65.0	(i) 70.1	(j) 80.8	(k) 68.4	(l) 77.7	(m) 66.7	(n) 75.1	(o) 7.4	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	99.4	74.6	64.9	94.3	73.1	63.4	89.1	71.8	34.4	80.8	33.0	77.9	31.6	75.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.0	(b) 17.7	(c) 15.8	(e) -23.2	(f) 88.0	(g) 7.5	(h) 3.3	(i) -28.6	(j) 90.4	(k) -32.9	(l) 92.4	(m) -37.1	(n) 94.2	(o) -42.6	(p) 96.7		
(a) 20.0	(b) 17.7	(c) 15.8	(e) -23.4	(f) 73.0	(g) 7.4	(h) 2.1	(i) -28.8	(j) 74.5	(k) -33.1	(l) 75.7	(m) -37.3	(n) 76.9	(o) -42.6	(p) 78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	11.8	13.7	24.8	39.7	53.1	60.3	65.0	61.0	50.9	38.1	25.8	16.4
	DBStd	21.06	13.13	12.47	8.37	9.09	9.14	7.90	6.80	7.49	7.71	7.89	10.08	12.96
	HDD50	5622	1183	1016	780	328	76	10	1	5	80	376	727	1040
	HDD65	9848	1648	1436	1245	758	380	183	85	171	426	834	1177	1505
	CDD50	1436	0	0	0	21	171	319	467	344	107	7	0	0
	CDD65	187	0	0	0	0	10	42	86	46	3	0	0	0
	CDH74	1551	0	0	0	0	117	367	701	344	22	0	0	0
	CDH80	355	0	0	0	0	17	77	167	92	2	0	0	0
Wind	WSAvg	7.6	8.6	8.9	8.8	8.1	7.6	6.7	5.8	6.2	6.6	7.7	8.2	8.7
Precipitation	PrecAvg	23.6	1.5	1.0	1.1	1.3	1.9	2.8	3.2	2.4	2.3	2.4	1.9	1.7
	PrecMax	30.7	3.0	2.0	3.3	4.4	4.2	4.9	9.5	5.4	4.1	7.2	3.9	2.6
	PrecMin	14.6	0.7	0.4	0.3	0.3	0.7	0.7	0.4	0.2	0.4	0.6	0.2	0.5
	PrecStd	3.7	0.6	0.5	0.6	0.9	0.9	1.2	2.4	1.3	1.1	1.3	0.8	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.9	47.4	71.0	82.4	86.6	89.1	88.0	76.8	63.3	48.3	40.6
		MCWB	33.3	33.7	39.9	54.1	64.1	69.3	69.9	69.8	61.1	52.8	46.2	40.3
	2%	DB	32.8	33.5	41.2	64.9	77.2	81.8	84.2	81.8	71.5	56.1	43.2	35.9
		MCWB	32.0	32.1	35.5	50.8	60.1	66.8	68.9	67.7	60.2	49.7	41.9	34.9
	5%	DB	31.6	32.2	38.1	58.9	72.8	78.2	81.1	77.4	67.1	52.0	39.9	33.3
		MCWB	31.0	31.4	33.8	47.9	58.0	65.4	67.5	65.7	57.9	48.3	38.8	32.4
	10%	DB	29.3	30.6	35.6	54.1	68.6	74.6	77.9	73.4	62.9	49.1	36.9	32.0
		MCWB	28.5	29.5	32.5	45.3	55.6	63.2	66.1	63.6	56.1	46.7	35.8	31.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	34.1	40.3	55.7	66.4	71.4	72.3	71.7	64.4	55.1	46.1	40.4
		MCDB	33.6	34.9	46.5	67.5	79.7	83.4	82.9	84.7	72.1	60.4	47.6	40.6
	2%	WB	32.3	32.5	36.6	52.1	63.0	69.1	70.3	69.3	61.1	51.0	42.2	35.2
		MCDB	32.7	33.2	39.7	61.9	74.0	79.2	80.9	79.4	68.7	54.4	42.9	35.8
	5%	WB	30.9	31.4	34.7	49.0	60.2	67.0	68.9	67.0	58.7	48.8	39.0	32.8
		MCDB	31.5	32.2	37.1	58.3	69.3	75.6	78.4	75.3	65.5	51.5	39.6	33.3
	10%	WB	28.5	29.5	32.9	45.7	57.4	64.8	67.3	64.9	56.5	46.7	36.0	31.5
		MCDB	29.2	30.6	35.1	53.1	66.2	72.4	75.8	72.0	62.5	49.0	36.6	32.0
Mean Daily Temperature Range	5% DB	MDBR	9.6	11.3	12.2	14.8	18.6	17.8	17.1	16.3	14.4	9.4	7.1	8.6
		MCDBR	7.5	8.6	13.3	21.4	23.9	22.1	21.5	21.1	20.9	14.7	8.6	6.1
		MCWBR	7.4	8.3	9.3	11.7	11.6	10.5	9.5	9.2	10.9	9.4	8.1	6.2
	5% WB	MCDBR	7.3	8.2	11.0	19.9	21.3	19.5	19.1	18.9	18.9	12.5	8.2	6.4
		MCWBR	7.5	8.1	8.4	11.7	11.7	10.5	9.3	9.2	10.5	9.2	8.0	6.6
Clear-Sky Solar Irradiance	taub	0.276	0.308	0.347	0.392	0.369	0.366	0.395	0.405	0.357	0.338	0.300	0.267	
	taud	2.063	2.063	2.031	2.130	2.375	2.413	2.345	2.297	2.415	2.347	2.186	2.027	
	Ebn at Noon	174	223	247	257	275	277	265	252	249	217	167	137	
	Edh at Noon	18	28	38	41	35	34	36	35	27	22	16	14	
All-Sky Solar Radiation	RadAvg	151	398	849	1262	1682	1792	1778	1358	803	350	128	84	
	RadStd	23	90	98	125	124	129	167	148	103	56	22	20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	N/A
(47) Regional (6 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	-277	N/A	N/A

Nomenclature: See separate page